

Solutions to MP363 Autumn Repeat Exam '12-'13

(1)

P.1

(a) This requires not only knowledge of Planck's blackbody radiation formula, but also how to approximate functions when their argument is small. For the former, they need to know that

$$I(\nu, T) = \frac{2\nu^2}{c^2} \frac{h\nu}{e^{h\nu/kT} - 1}$$

[14pts]

was Planck's modified intensity formula. To find the leading-order behavior for this for small ν (or, more correctly, small $h\nu/kT$), you can either recall the series formula for the exponential or Taylor expand this around $\nu=0$. The former is easier, so let's do that: $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$

$$\text{so } e^{h\nu/kT} - 1 = \frac{h\nu}{kT} + \frac{1}{2} \left(\frac{h\nu}{kT} \right)^2 + \frac{1}{6} \left(\frac{h\nu}{kT} \right)^3$$

$$\approx \frac{h\nu}{kT}$$

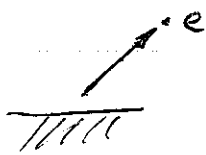
for small ν , because each successive term is $\frac{h\nu}{kT}$ smaller than the preceding. Thus:

$$I(\nu, T) \approx \frac{2\nu^2}{c^2} \frac{h\nu}{h\nu/kT} = \frac{2\nu^2 kT}{c^2}$$

[12pts]

which is precisely the Rayleigh-Jeans formula. So for small frequencies, they agree, but they certainly do not for large ν .

(b) The key to this one is remembering how the photoelectric effect works: a photon



of energy E_γ hits a substrate; in order to

penetrate and kick an electron out, it must overcome a potential barrier, the "work function" W . The remaining energy goes into the kinetic energy of the electron. So conservation of

energy tells us that $E_\gamma = E_e + W$. Since $E_\gamma = h\nu = hc/\lambda$ for a photon, we have the rel'n

$$\boxed{\frac{hc}{\lambda} - W = E_e}$$

So if we have two values for the pair (λ, E_e) , we can find h and W : $\frac{hc}{\lambda_1} - W = E_{e1}$, $\frac{hc}{\lambda_2} - W = E_{e2} \Rightarrow$

$$h = \frac{E_{e1} - E_{e2}}{c(\frac{1}{\lambda_1} - \frac{1}{\lambda_2})}, \quad W = \frac{E_{e1}\lambda_1 - E_{e2}\lambda_2}{\lambda_2 - \lambda_1}$$

We're told that a photon of wavelength $\lambda_1 = 3 \times 10^{-7} \text{ m}$ kicks out an electron with $E_{e1} = 1.4 \text{ eV}$, and one with $\lambda_2 = 4 \times 10^{-7} \text{ m}$ gives $E_{e2} = 0.4 \text{ eV}$. Putting these numbers in gives

$$\boxed{h = 4 \times 10^{-15} \text{ eV} \cdot \text{s} = 6.4 \times 10^{-34} \text{ J} \cdot \text{s}}$$

[12pts]

$$\boxed{W = 2.6 \text{ eV}}$$

[12pts]

(The correct values are $6.6 \times 10^{-34} \text{ J} \cdot \text{s}$ and 2.36 eV , so these are pretty close.)

P.2

(a) This requires students to remember the definition of an expectation value, but also needs them to know what a Hermitian operator is and how to use the inner product correctly.

First, the def'n: if A is an operator on $\mathcal{H}$ and ψ a state in $\mathcal{H}$, then the expectation value of A for this state, $\langle A \rangle_\psi$, is defined to be

$$\boxed{\langle A \rangle_\psi := \langle \psi | A \psi \rangle}$$

[10pts]

i.e., the inner product of ψ with the state $A\psi$.

Now, what's the complex conjugate of $\langle A \rangle_\psi$? Since we

know $\langle \psi | \phi \rangle = \overline{\langle \phi | \psi \rangle}$, we see that

$$\begin{aligned} \langle A \rangle_\psi &= \langle \psi | A \psi \rangle = \overline{\langle A \psi | \psi \rangle} \\ &= \langle \psi | A^\dagger \psi \rangle \end{aligned}$$

since the def'n of $A^\dagger$ is the operator such that

$$\langle \psi | A^\dagger \phi \rangle = \langle A \psi | \phi \rangle$$

for all $\psi, \phi \in \mathcal{H}$. So $\langle A \rangle_\psi = \langle A^\dagger \rangle_\psi$, the expectation value of $A^\dagger$. However, if A is a Hermitian operator, then

(3)

by defn $A^\dagger = A$, so this implies $\overline{\langle A \rangle_\psi} = \langle A \rangle_\psi$. And the only numbers that are unchanged by complex conjugation are real numbers. Thus, if A is Hermitian, then $\langle A \rangle_\psi$ is real. [15pts]

(b) There are several ways to do this problem, some more abstract than others. We'll do it two ways:

Abstract - $\langle \psi | \psi \rangle$, in principle, is a function of t , because $\psi(t)$ is a time-dependent soln to $H\psi = i\hbar \frac{\partial \psi}{\partial t}$. Now, we can take its time derivative:

$$\frac{d}{dt} \langle \psi | \psi \rangle = \langle \frac{\partial \psi}{\partial t} | \psi \rangle + \langle \psi | \frac{\partial \psi}{\partial t} \rangle$$

because $\langle \psi | \psi \rangle$ is a product of $\langle \psi |$ and $|\psi\rangle$, so we use the usual product rule for derivatives. But $\frac{\partial \psi}{\partial t} = -\frac{i}{\hbar} H\psi$ from Schrödinger's eq'n, so

$$\frac{d}{dt} \langle \psi | \psi \rangle = \langle -\frac{i}{\hbar} H\psi | \psi \rangle + \langle \psi | -\frac{i}{\hbar} H\psi \rangle$$

$$= \frac{i}{\hbar} \langle H\psi | \psi \rangle - \frac{i}{\hbar} \langle \psi | H\psi \rangle$$

because $\langle \alpha\psi | \phi \rangle = \alpha \langle \psi | \phi \rangle$ and $\langle \psi | \alpha\phi \rangle = \alpha \langle \psi | \phi \rangle$ for $\alpha \in \mathbb{C}$. $\langle H\psi | \psi \rangle = \langle \psi | H^\dagger \psi \rangle$ (see part a), but the Hamiltonian, being an observable, is Hermitian, so $H^\dagger = H$.

Thus,

$$\boxed{\frac{d}{dt} \langle \psi | \psi \rangle = \frac{i}{\hbar} \langle \psi | H\psi \rangle - \frac{i}{\hbar} \langle \psi | H\psi \rangle = 0.}$$

[15pts]

Less abstract - here we use the explicit formula for the inner product on wave functions:

$$\langle \psi | \psi \rangle = \int_{-\infty}^{\infty} \overline{\psi(x,t)} \psi(x,t) dx$$

(where we have assumed a 1-dimensional system for simplicity).

Then,

$$\frac{d}{dt} \langle \psi | \psi \rangle = \int_{-\infty}^{\infty} \frac{\partial}{\partial t} [\overline{\psi(x,t)} \psi(x,t)] dx$$

$$= \int_{-\infty}^{\infty} \left(\frac{\partial \bar{\psi}}{\partial t} \psi + \bar{\psi} \frac{\partial \psi}{\partial t} \right) dx$$

$$= \int_{-\infty}^{\infty} \left(\frac{\partial \bar{\psi}}{\partial t} \psi + \bar{\psi} \frac{\partial \psi}{\partial t} \right) dx$$

because t is real. Now, we once again use $\frac{\partial \psi}{\partial t} = -\frac{i}{\hbar} H\psi$ to get

$$\begin{aligned}
 \frac{d}{dt} \langle \psi | \psi \rangle &= \int_{-\infty}^{\infty} \left[\overline{\left(-\frac{i}{\hbar} H \psi \right)} \psi + \bar{\psi} \left(-\frac{i}{\hbar} H \psi \right) \right] dx \\
 &= \int_{-\infty}^{\infty} \left[\frac{i}{\hbar} \overline{(H \psi)} \psi - \frac{i}{\hbar} \bar{\psi} (H \psi) \right] dx \\
 &= \frac{i}{\hbar} \int_{-\infty}^{\infty} \overline{(H \psi)} \psi dx - \frac{i}{\hbar} \int_{-\infty}^{\infty} \bar{\psi} (H \psi) dx \\
 &= \frac{i}{\hbar} \langle H \psi | \psi \rangle - \frac{i}{\hbar} \langle \psi | H \psi \rangle
 \end{aligned}$$

and proceed as before i.e., $H^\dagger = H \Rightarrow \frac{d}{dt} \langle \psi | \psi \rangle = 0$.

Either way, we conclude that $\langle \psi | \psi \rangle$ is independent of time, and thus if it's set equal to 1 at any time t_0 ,

it's 1 at all times t . [w. that this property, $\langle \psi | \psi \rangle$ could be something other than 1 at some time other than t_0 .] [10pts]

P.3

This problem deals with how to write the coefficients in the expansion of a state in terms of an orthonormal basis of energy eigenstates. The only real maths that's required is knowing that $\sum_{n=0}^{\infty} \alpha r^n = \frac{\alpha}{1-r}$ if $|r| < 1$, which (hopefully) the students should know by heart...

(a) The state given has the form $\Psi = \sum_{n=0}^{\infty} c_n \psi_n$, where $c_n = \frac{A}{3^n}$. In lecture and on the problem sheets, we learned that the condition $\langle \Psi | \Psi \rangle = 1$ is equivalent to the condition $\sum_{n=0}^{\infty} |c_n|^2 = 1$, so this should allow us to determine A .

$$\begin{aligned}
 \sum_{n=0}^{\infty} |c_n|^2 &= \sum_{n=0}^{\infty} \left(\frac{A}{3^n} \right)^2 = A^2 \sum_{n=0}^{\infty} \left(\frac{1}{9} \right)^n = A^2 \frac{1}{1 - 1/9} \\
 &= \frac{9A^2}{8}
 \end{aligned}$$

Now, if A is a positive real number such that $9A^2/8 = 1$, this

gives

$$\boxed{A = 2\sqrt{2}/3}$$

[15pts]

(5)

(b) One of the basic postulates of QM states that if a state has the form $\Psi = \sum_n c_n \psi_n$, where $\{\psi_n\}$ is a complete set of orthonormal energy eigenstates, then the probability that a measurement of the energy gives the specific eigenvalue E_n is $|c_n|^2$. We've told here that the energy eigenvalues are $E_n = -E/2^n$, so the possible energies are $E_0 = -E$, $E_1 = -E/2$, $E_2 = -E/4$, etc. So if we measure $-E/2$, that tells us that we're getting the energy of the $n=1$ state, so the probability is

$$|c_1|^2 = \left| \frac{A}{3^1} \right|^2 = \left(\frac{2\sqrt{2}}{9} \right)^2 = \boxed{\frac{8}{81} = 0.098765432} \quad [15 \text{ pts}]$$

or, we set $-E/2$ a bit less than 10% of the time.

(c) Finally, the utility of the expansion gives a specific form for the expectation value $\langle H \rangle_\Psi$ of the energy:

$$\langle H \rangle_\Psi = \sum_n |c_n|^2 E_n.$$

So, for the case here, we have

$$\begin{aligned} \langle H \rangle_\Psi &= \sum_{n=0}^{\infty} \left| \frac{A}{3^n} \right|^2 \left(-\frac{E}{2^n} \right) = -A^2 E \sum_{n=0}^{\infty} \left(\frac{1}{18} \right)^n \\ &= \left(-\frac{8E}{9} \right) \left(\frac{1}{1 - 1/18} \right) = -\left(\frac{8E}{9} \right) \left(\frac{18}{17} \right) = \boxed{-\frac{16E}{17}} \quad [20 \text{ pts}] \end{aligned}$$

as expected. (This also gives the students a chance to see if they got the right value for A in (a); if not, they'll get a different answer for $\langle H \rangle_\Psi$.)